

1. Fight bout trajectory data

$$x = (D_{PP}, \theta_W, \theta_L)$$

2. Exploratory SFI fit

325 basis functions per force

3. Inspect inferred structure

force field heatmaps and dynamics

4. Construct shared reduced basis

39 interpretable basis functions

5. Final SFI fit

shared basis for F_D , F_{θ_W} , and F_{θ_L}

6a. Model validation

stochastic simulation
Jensen-Shannon distance
residual diagnostics

6b. Simplification and interpretation per force

identify dominant terms
 F_D : constant + cosine terms
 $F_{\theta_W}, F_{\theta_L}$: turning + gated terms